Postprandial lipaemia: effects of sitting, standing and walking in healthy normolipidaemic humans.

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Long periods of sedentary behaviour may adversely affect health irrespective of overall physical activity levels. This study compared the effects of sitting, standing and walking on postprandial lipaemia in healthy normolipidaemic Japanese men. 15 participants, aged 26.8±2.0 years (mean±SD), completed 3, 2-day trials in a random order: 1) sitting (control), 2) standing, and 3) walking. On day 1 of the sitting trial, participants rested. On day 1 of the standing trial, participants stood for six, 45-min periods. On day 1 of the walking trial, participants walked briskly for 30 min at approximately 60% of maximum heart rate. On day 2 of each trial, participants rested and consumed test meals for breakfast and lunch. Venous blood samples were collected in the morning and afternoon on day 1, and in the fasted state (0 h) and at 2, 4 and 6 h postprandially on day 2. On day 2 area under the serum triacylglycerol concentration vs. time curve was 18% lower on the walking trial than the sitting and standing trials (1-factor ANOVA, $P=0.015$). Hence postprandial lipaemia was not reduced after standing but was reduced after low-volume walking compared with sitting in healthy normolipidaemic Japanese men.